Real-World Order of Operations

Question Bank 4.07

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $4x^2$ when $x = -3$

2. $\frac{1}{2}mv^2$ when $m = 50$ and $v = 6$

3. $\frac{a+b^2}{c}$ when $a = 6$, $b = 5$, $c = 2$

4. $2(x-y)^2$ when $x = 3$ and $y = 7$

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 150$, $h = 65$

b. $w = 180$, $h = 70$

c. $w = 200$, $h = 72$

8. Kinetic energy is $K=\frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m=1{,}200$, $v=15$

b. $m=850$, $v=12$

c. $m=70$, $v=8$

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2=900$ and $v=6$. Find m.

b. $x^2+5=41$, where x is positive. Find x.

c. A square lawn of side n feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find n.

Reading the situation.

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.

11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?

- ○ A. $15(4.80)+120$
- ○ B. $15^2(4.80)+120$
- ○ C. $15^2(4.80+120)$
- ○ D. $15^2+4.80+120$

12. Select ALL of the expressions whose value is 45 when $x=3$.

- ☐ A. $(5x)^2$
- ☐ B. $\frac{135}{x}$
- ☐ C. $5x^2$
- ☐ D. $3x^2+18$
- ☐ E. $5x\cdot 2$
- ☐ F. x^2+36